

# RACHIT KALPESHKUMAR SHAH

CONTACT NO.: (+91) 9408784649 || E-MAIL: [shahrachit0202@gmail.com](mailto:shahrachit0202@gmail.com)

Cyber Security professional with hands-on experience in Vulnerability Assessment & Penetration Testing (VAPT), Red Team Operations, SOC Monitoring, and ICS/SCADA Security. Experienced in conducting web and network security assessments, vulnerability exploitation, security reporting, and threat analysis through internships at WESEE (Indian Navy), CiberTroops Security Solutions, and participation in national cyber exercises. Author of Springer research publications focused on network security, phishing detection, and deepfake analysis.

## EDUCATION

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### Master of Technology in Cyber Security

July 2024 – May 2026

Rashtriya Raksha University, Lavad, Gujarat

- Pursuing MTech in Cyber Security with focus on network defense, vulnerability assessment and secure system design.

### Bachelors of Information and Technology

July 2020 – May 2024

Charotar University of Science and Technology, Changa, Gujarat

- Bachelor's in information technology with foundation in software systems, networking, databases, and core computing principles.

## INTERNSHIP

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### VAPT Intern

Oct 2025 – March 2026

Weapons and Electronics Systems Engineering Establishment (WESEE), Indian NAVY, Delhi

- Led hands-on Vulnerability Assessment and Penetration Testing (VAPT) using Nmap, Burp Suite, Nessus, and Metasploit to uncover and validate exploitable weaknesses across web and network environments.
- Drove reconnaissance, attack-surface mapping, and web application exploitation aligned with OWASP Top 10, prioritizing risks using CVSS scoring and threat impact analysis.
- Delivered high-quality technical reports containing PoC demonstrations, root-cause analysis, and actionable remediation strategies, improving security posture visibility for engineering teams.
- Contributed to cybersecurity evaluation activities within a defense R&D setting, strengthening secure development and testing practices for mission-critical systems.

### VAPT Intern

Nov 2024 – March 2025

CiberTroops Security Solutions, Surat

- Performed vulnerability assessments on web applications and network infrastructures using tools like Nessus, OpenVAS, and Nikto, identifying critical security gaps and misconfigurations.
- Executed penetration testing methodologies including OWASP Top 10 exploitation, SQL injection, XSS, and authentication bypass, developing proof-of-concept exploits to demonstrate impact.
- Compiled comprehensive technical reports detailing findings, risk ratings, and remediation recommendations in accordance with industry best practices and client security policies.

### Cyber security & Machine learning Intern

Jan 2024 – April 2024

Charusat University, Anand

- Conducted research on detecting deepfakes in image and video files through the project “DeepVerify: Ensuring Authenticity through Deepfake and Liveness Analysis.”
- Explored and analyzed CNN-based machine learning and deep learning approaches for identifying deepfakes in multimedia content.
- Investigated the role of Generative Adversarial Networks (GANs) in deepfake creation to enhance detection accuracy and robustness.
- Presented findings by building an experimental system with a Streamlit interface, demonstrating practical applications of the research.
- Conference: 9th International conference on ICT for sustainable development

### **SOC analyst Intern**

**Sept 2022 – Jan 2023**

Elastic run, Pune

- Monitored and analyzed network traffic in a Security Operations Center environment, utilizing Trend Micro File Storage Security, Endpoint Workload Security, and Container Security integrated with Microsoft Azure.
- Detected and reported suspicious activities and malware, enabling preventive measures that reduced potential security breach risks by approximately 30 percent.
- Applied expertise in cloud security management, incident response, and threat detection to strengthen organizational security posture.

## **PROJECT DETAILS**

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### **CcureGen: Data securing android application**

Technology: Android Development || Cyber Security

- CcureGen is a security application designed to protect data using three key features: Snort, Hash calculation, and Steganography. Snort filters packets and detects intrusions on the network.
- Whereas, hash verifies the integrity of uploaded media. And, steganography detects malware hidden within media files, ensuring comprehensive data security.

## **ACHIEVEMENTS**

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|-----------------------------------------------------------------------------------------------|-----------------------------------------------------|
| ● Data Base Management System – NPTEL                                                         | ● Jr Penetration Tester - TryHackMe                 |
| ● Introduction to Database management System – NPTEL                                          | ● Ethical Hacking Essentials –CodeRed, EC Council   |
| ● Intro to DFIR: the divide and conquer process Basis Technology                              | ● Azure Fundamentals (AZ-900) – Microsoft           |
| ● Introduction to Cyber Security – CISCO                                                      | ● Blockchain Foundation program – KBA               |
| ● Google Cloud third milestone achieved in the Google Cloud Readiness Program 2022, 58 badges | ● Ethereum Foundation Program – KBA                 |
| ● Web Fundamentals - TryHackMe                                                                | ● Cloud Security Badge – AWS                        |
|                                                                                               | ● Google Cloud Career Readiness Program 2023–Google |

## LEADERSHIP EXPERIENCE

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### CII SECEX

April-2025

#### Red Team member

Delhi

Participated in a live fire cyber exercise involving Red Team attacks, Blue Team defense, and White Team evaluation of incident response.

- Conducted reconnaissance, vulnerability assessment, exploitation, and advanced adversary tactics including privilege escalation, lateral movement, and persistence.
- Produced technical evidence and incident reports for White Team review, enhancing skills in offensive security and structured documentation.

### BHARAT NCX

July 2025

#### ICS Security Trainer Assistant / Red Team member

Delhi

- Supported live training sessions for government, private, and public sector delegates on ICS/SCADA security operations and protocol analysis.
- Configured Snort Intrusion Detection System with custom rules to detect malicious Modbus TCP traffic and unauthorized control operations.
- Demonstrated ICS exploitation techniques and guided implementation of defensive strategies including network segmentation, intrusion detection, and OT-specific risk mitigation.

## RESEARCH WORK

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### Smishing: A SMS Phishing Detection Using Various Machine Learning Algorithms

Technology Stack: Python || Machine Learning || Cyber Security

Conference: 5th international conference on Communication and Intelligent Systems; Springer.

- Designed and implemented a machine learning-driven application for detecting phishing attacks in SMS by analyzing both header and message content, along with integrated URL inspection to ensure authenticity and enhance mobile communication security.

### Enhancing Security in a University Network Using Snort and Wireshark

Technology Stack: Snort || Wireshark || Cyber Security

Conference: 8th international Conference on ICT for Intelligent Systems; Springer

- The research aims to improve university network security by monitoring suspicious activities and utilizing Snort as a Network Intrusion Detection System (NIDS) for threat response, while Wireshark for network data capture and analysis, including dark web monitoring

## SKILLS

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- **Platforms:** Windows, Kali Linux, Parrot OS, MAC OS, MS-Office suite
- **Networking & Security:** Wireshark, Snort, Nmap, Burp Suite, Security suite, Splunk, Sliver, Havoc, SQLmap
- **Programming:** Java (Beginner), Javascript, Python (Beginner), Snort, SQL, Bash, Powershell
- **Cloud & Infrastructure Platforms:** Oracle, AWS, Azure, Google Cloud
- **Language:** Gujarati (Mother tongue), English (Intermediate), Hindi (Intermediate)